Pre-Reads: Git & GitHub - Basics

Nishut Suman 31 Dec, 2025 At 10:00 AM Pre-Reads Foundation Recommended Resource

Associated Lectures (1)

Description Discussions

Pre-Read: Git & GitHub

⭐ What You'll Gain from This Pre-Read

After reading, you'll be able to:

- Understand what **version control** is and why every developer uses it.
- Differentiate between **Git (the tool)** and **GitHub (the cloud platform)**.
- Install **Git** on your computer (Windows or macOS).
- Configure Git using your **name and email**, so your commits are identified properly.
- Understand the **3-stage Git workflow** - working directory, staging area, and repository.
- Know the **essential commands** to start your first project (**git init, git add, git commit**).

Think of this as:
Learning how to *save your project history forever* - so you can travel back in time when something goes wrong.

📘 What This Pre-Read Covers

- Why version control matters
- What Git and GitHub actually are
- How to install Git (Windows + macOS)
- Initial configuration (**git config**)
- Core Git workflow and basic commands
- How to start your first repository

🚀 Part 1: The Big Picture - Why Does This Matter?

> 💬 "Have you ever broken something in your code and wished you could undo it?"

Git is that **magical "undo" button** - but not just for one file. It tracks every change across your entire project, from start to finish.

GitHub then takes this one step further - it's the **cloud home** for your code. You can share, collaborate, and build amazing things with your team or community.

🧑‍💻 Where You'll Use This

Job Roles:

- **Software Developers:** Track versions and experiment safely.
- **Data Scientists:** Reproduce and version their experiments.
- **DevOps Engineers:** Automate deployments from Git repositories.

Real Products Using GitHub:

- Google and NASA use Git for massive projects.
- Netflix, Spotify, and Airbnb rely on GitHub for team collaboration.

What You Can Build:

- Personal or academic projects
- Open-source contributions
- Full production-grade systems

🧩 Think of It Like This

Git is your **"time machine" for code** - it records every change. GitHub is your **"Google Drive" for code** - it stores and shares those records.

⚠️ **Limitation:**
Git doesn't fix broken logic - it simply keeps a history so you can fix it confidently.".

🧭 Part 2: Installing Git (Step-by-Step)

Let's set up your Git environment before class!

🪟 For Windows Users

Step 1: Download Git

Go to https://git-scm.com/downloads

You'll see something like:

Click **“Download for Windows”**. It automatically picks the right version (64-bit).

Step 2: Run the Installer

Double-click the downloaded file (**Git-2.x.x-64-bit.exe**).
Keep the **default settings** for everything (press *Next* repeatedly).

When you reach:

- **Editor:** Choose *Visual Studio Code* or *Notepad* (if unsure).
- **PATH Environment:** Select “Git from the command line and also from 3rd-party software.”

Click **Install**, then **Finish**.

Step 3: Verify Installation

Open **Command Prompt** or **Git Bash** and type:

```
git --version
```

✅ You should see something like:

```
git version 2.45.0.windows.1
```

🍏 For macOS Users

Option 1: Use Homebrew (Recommended)

If you have **Homebrew**, just run:

```
brew install git
```

Option 2: Manual Download

Go to https://git-scm.com/download/mac\ You’ll see something like:

Download the **.dmg** installer and drag the Git icon into Applications.

Step 3: Verify Installation

Open **Terminal** and type:

```
git --version
```

You should see something like:

```
git version 2.45.0
```

⚙️ Part 3: Initial Git Configuration

After installation, you need to introduce yourself to Git - this identity will appear in all your commits.

```
# Set your name
git config --global user.name "Your Name"

# Set your email
git config --global user.email "youremail@example.com"
```

👉 These commands tell Git who you are when you save changes.

You can check your configuration anytime with:

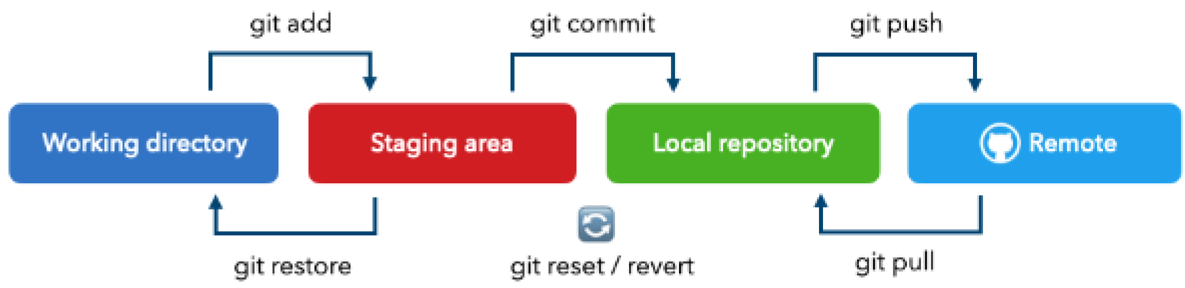
```
git config --list
```

Example Output:

```
user.name=alex
user.email=alex@example.com
```

> 💡 Tip: Use the same email you plan to sign up with on GitHub - it keeps things consistent.

🗂️ Part 4: The 3-Stage Git Workflow



Git has **three** main zones that manage your files.:

Zone	Description	Example Command
Working Directory	Where your normal files live	—
Staging Area	Temporary holding area for changes	**git add .** (Remember to leave a space after add before the .)
Repository	Permanent history (commits)	**git commit -m "message"**

This workflow ensures you **review changes before saving history**.

🧩 Part 5: Your First Repository

Let's practice - your "Hello Git!" moment.

```
# Step 1: Create a new folder
mkdir my-first-repo
cd my-first-repo

# Step 2: Initialize Git
git init

# Step 3: Create a sample file
echo "print('Hello Git!')" > hello.py

# Step 4: Add to staging
git add hello.py

# Step 5: Commit changes
git commit -m "Initial commit"

# Step 6: Check history
git log
```

✅ You've now created your **first Git repository!** Git is officially tracking your project from this point forward.

🧩 Part 6: .gitignore - Keeping Your Repo Clean

Sometimes you don't want Git to track everything - like secrets, passwords, or temporary files.

Create a file named **.gitignore** in your project folder:

```
# .gitignore
__pycache__/
*.log
.env
```

This tells Git to *ignore* these files completely.

🌍 Part 7: GitHub - Where Your Code Lives Online

Now that you've set up Git locally, it's time to connect to the cloud.

- **Git:** Your local tool for tracking versions.
- **GitHub:** A platform to host, share, and collaborate on those versions.

You'll soon learn to:

- Create a GitHub account
- Upload (push) your local repo
- Pull changes from collaborators

📘 Recommended Visual & Reading Resources

- 📘 Pro Git (Official Free Book)
- 🎥 Traversy Media – Git & GitHub Crash Course (YouTube)
- 🎨 Git Visualization Tool (learngitbranching.js.org) – See Git commands visually (**Must Try)

🌈 Final Thought

Every professional developer - from your favorite open-source contributors to engineers at NASA - started by running **git init**.

Today, **you've just learned how to begin your version control journey** - your future projects will thank you for it.